AMIT VERMA

ML Engineer · Python Developer · Computer Vision · NLP · LLMs

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PROFILE

Final-year CS undergraduate with DRDO (SSPL) research experience and a peer-reviewed publication at ICSSAI 2025. Specialises in LLM fine-tuning, RAG pipelines, transformer-based NLP, medical imaging AI, and computer vision. Built local ML systems with memory support, leveraging mixed-precision training, FAISS/ChromaDB vector retrieval, and local LLM deployment (Llama 3.1, Qwen 2.5) via llama-cpp on hardware.

TECHNICAL SKILLS

Languages
Python (Advanced) · C++ · JavaScript

ML / Deep Learning
PyTorch · Transformers · scikit-learn · Multimodal Models

Computer Vision
OpenCV · GradCAM / GradCAM++ · Thermal Imaging · Landmark Detection

Python Ecosystem
Flask · FastAPI · NumPy · Pandas · Matplotlib · PyQt

Tools & Infra
Git · Linux · REST APIs

EDUCATION

B.Tech in Computer Science and Business Systems

Techno India University, Kolkata Aug 2022 – Apr 2026
CGPA: **8.1 / 10**

LANGUAGES

English ▪ **Hindi** ▪ **Bengali**

RESEARCH & ACHIEVEMENTS

Publication — ICSSAI 2025

Comparative study of image denoising under Gaussian noise; Group Normalization achieved **7.9% higher PSNR** vs. Batch Normalization. Accepted & presented at *ICSSAI 2025*.

Amazon ML Challenge

End-to-end **multimodal pipeline** for product attribute prediction on **2M+ samples**: MiniLM text encoding and EfficientNet-B0 image features fused into a shared embedding space; deep MLP head with BatchNorm & Dropout for robust regression to predict price from description and images.
Rank 1570 · **Score 51.20** · GitHub

EXPERIENCE

Software Engineering Intern
Feb -- May 2025
TIU – DRDO Executive Lab

- Built an **image processing application** with **50+ filters** and integrated **N2N, N2V, DnCNN**, improving **PSNR/SSIM by 10%**.
- Developed a **multithreaded video pipeline** with real-time controls, reducing latency by **35%**.
- Processed **5D thermal (.mat) data** using **NUC** to generate smooth thermal videos.
- Fixed crashes and memory leaks, improving system stability.

PROJECTS

1. Local LLM Memory System with RAG
GitHub
Llama 3.1 8B · Qwen 2.5 14B · llama-cpp · ChromaDB · sentence-transformers · Python

- Deployed Llama 3.1 8B and Qwen 2.5 (14B/7B) locally via llama-cpp; optimized inference using layer-wise offloading and built a lightweight desktop UI with **Tauri**.
- Designed persistent memory using **ChromaDB** with semantic retrieval across sessions.
- Implemented modular **RAG pipeline** for context-aware long-horizon conversations and document querying.

2. Multitask Sentiment Analysis with Sarcasm Detection
GitHub Demo
XLM-RoBERTa · HuggingFace · Multitask Learning · FAISS · Mixed Precision

- Achieved **93% validation accuracy** on **200k+ samples** using multitask learning with sarcasm-aware gating.
- Built **3 FAISS sub-indexes** to eliminate cross-polarity noise in semantic retrieval.
- Stabilized training by resolving NaN issues via adaptive gradient clipping and class-weighted loss.

3. Brain Tumor Classification with GradCAM Explainability
GitHub Demo
PyTorch · CNN · GradCAM · GradCAM++ · Data Augmentation

- Developed a deep learning pipeline achieving **91% accuracy** on **4,200+ MRI scans** across **4** tumor classes.
- Integrated **GradCAM and GradCAM++** to generate interpretable heatmaps for clinical validation.
- Engineered a hybrid **EfficientNet-B0 + Swin Transformer** model with attention-based feature fusion.
- Applied advanced augmentation and regularization to improve generalization on medical imaging data.

4. AIMLverse — Interactive ML Education Platform
Demo
Next.js 14 · Three.js · Python · Interactive Visualisation

- Built and deployed an interactive platform with **3D visualizations** for LLMs, CNNs, and MLPs.
- Developed real-time demos for backpropagation, normalization, and decision boundaries using Three.js.
- Integrated custom Python backends for dynamic ML simulations.

5. Face Cursor Control via Facial Landmark Detection
GitHub
MediaPipe · OpenCV · Python Multithreading · Real-time CV

- Built an accessibility tool enabling webcam-based cursor control using facial landmarks and wink detection.
- Achieved **40% performance improvement** using multithreading for real-time responsiveness.